

Lecture 30

- Let's spend the rest of the course talking about interesting things 😊
- things that you probably won't see in any other course at CNU, which is just wrong, IMNSHO.

TOPIC 1

Relativity and Albert Einstein

Einstein was incredibly brilliant, and had a lot of personal flaws and issues in his life. He

and major issues ...
is a fascinating person; you should
read about him.

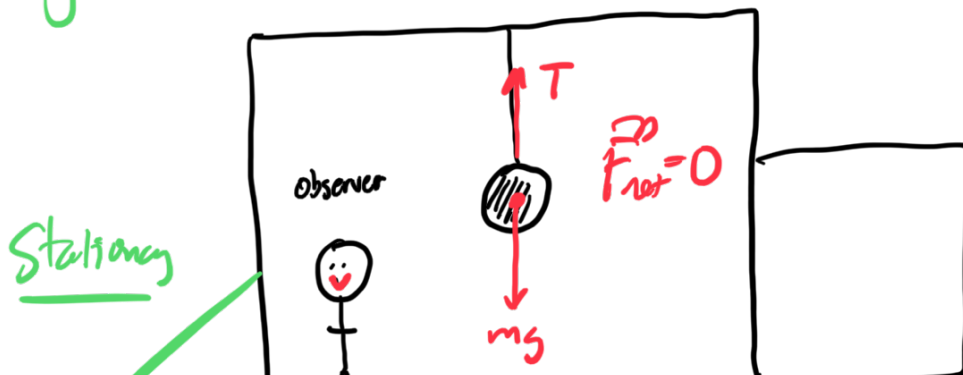
- ① He asked hard and
probing questions.
- ② He was willing to go
against the commonly accepted
explanation of things.

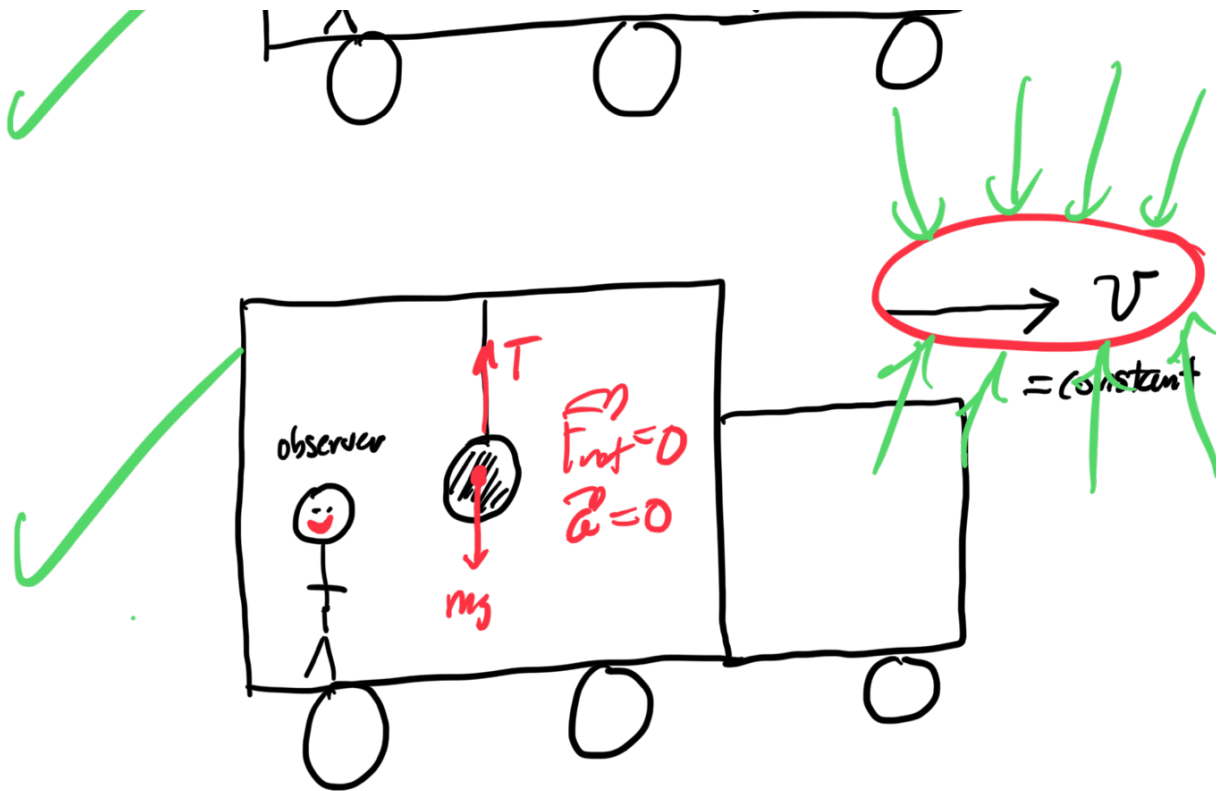
Newton's Laws, and Accelerating
Reference Frames

Gedanken experiments

$$\vec{F}_{\text{net}} = 0$$
$$= m\vec{a}$$

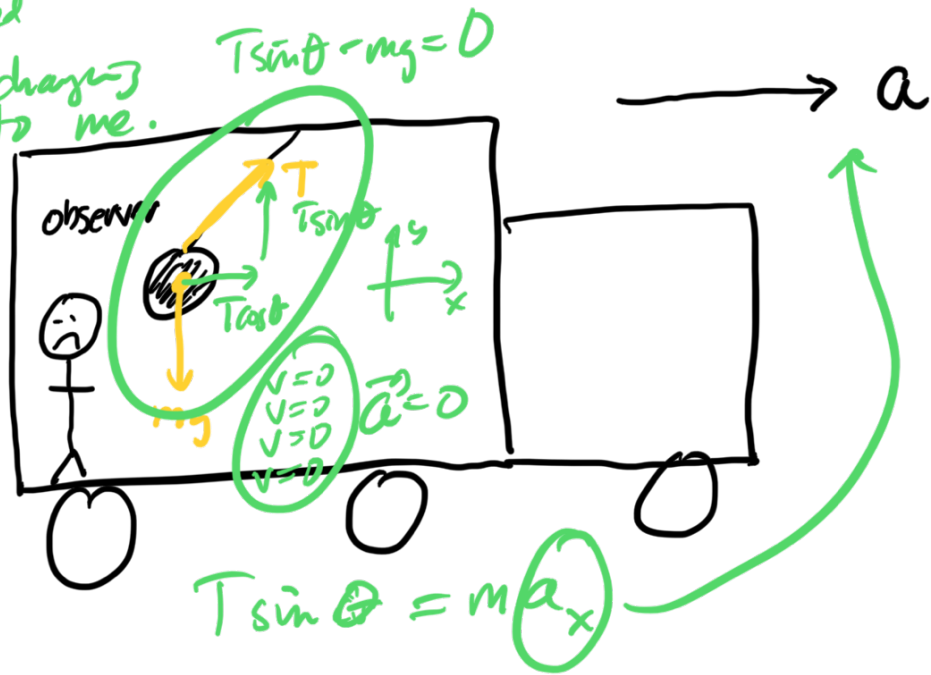
constant velocity





"It's speed is not changing relative to me."

"I'm so confused... there are magical forces in my universe"



Conclusions:

① Newton's Laws only work in non-accelerating

reference frames.

- ② The speed of the reference frame can never appear in the equations of motion.

BIG PROBLEM !

Maxwell's Equations → describe EM phenomena, and light!

→ the speed of light appears in the equations of motion.



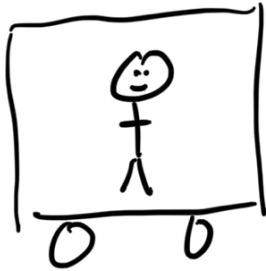
v_{light}

$= c$

$(\sim 3 \times 10^8 \text{ m/s})$

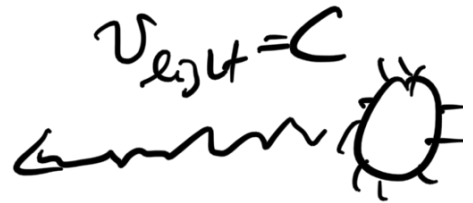


$$(2.997 \times 10^8 \text{ m/s})$$

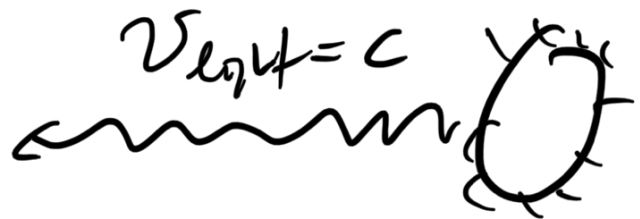


v_{truck}

relative to him



$$v_{\text{light}} = c$$



v_{truck}

$$v_{\text{light}}^{\text{relative to him}} =$$

$$= v_{\text{light}}^{\text{source}} + v_{\text{truck}}$$

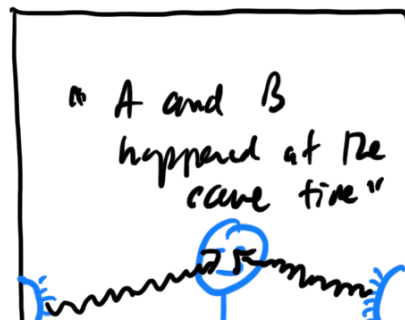
~~$$= c + v_{\text{truck}}$$~~

$$= c$$

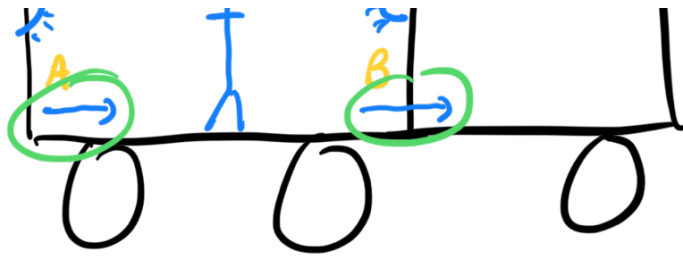
Huh?

"The speed of light is the same in all reference frames."

How?



$$v = 0.8c$$

→ light A and light B flash at the same time.

light
waves
are
moving.

outside
the train.

$v = 0.8c$

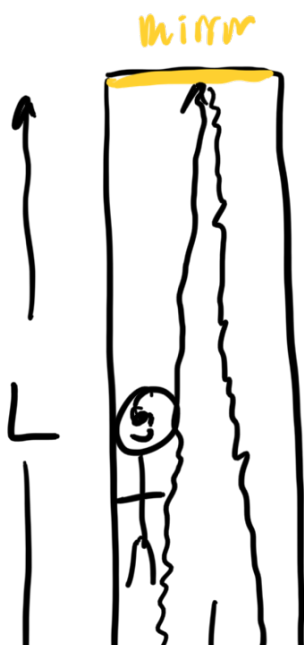
light from A and B do not
reach his eyes at the same
time.

"A and B happened at different times"

Principle of Simultaneity.

"Our perception of time depends on whether we are moving relative to the event that we are observing"

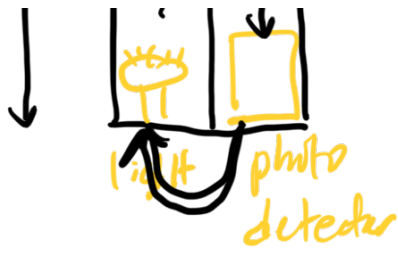
Build a Clock



Earth

1 year older.

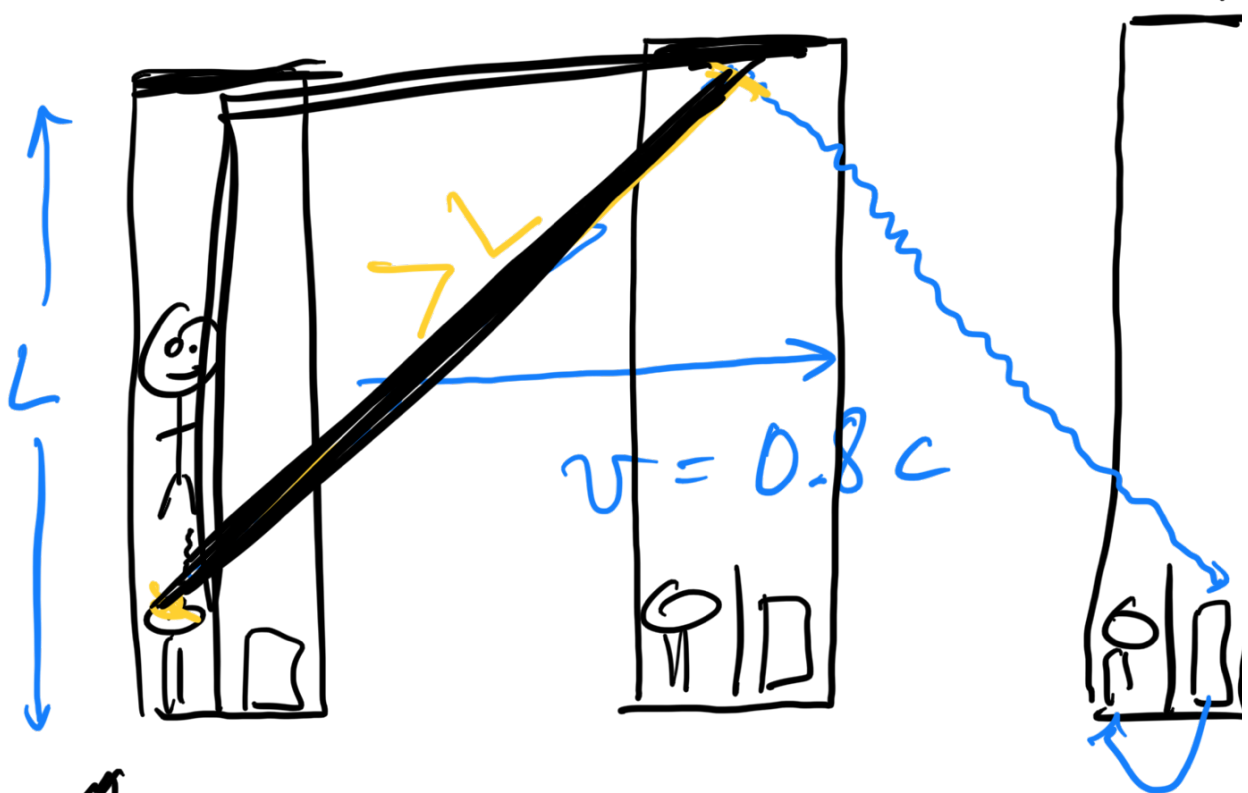
... 107 ticks



$$3.15 \times 10^7$$

==

design: L such
that light
flashes 1 time
per second.



Moving Clock run
class 1

slower
— ticks are every 1.7 seconds.
" $\frac{1}{1.7}$ of a year " older.

"TWIN PARADOX"

Space and time are
not separate dimensions

Time Dilation



1 1 1

dilation factor.

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Jet Airplane
($v = 600 \text{ mph}$)

$$\gamma = 1.0000007$$

Atomic Clock



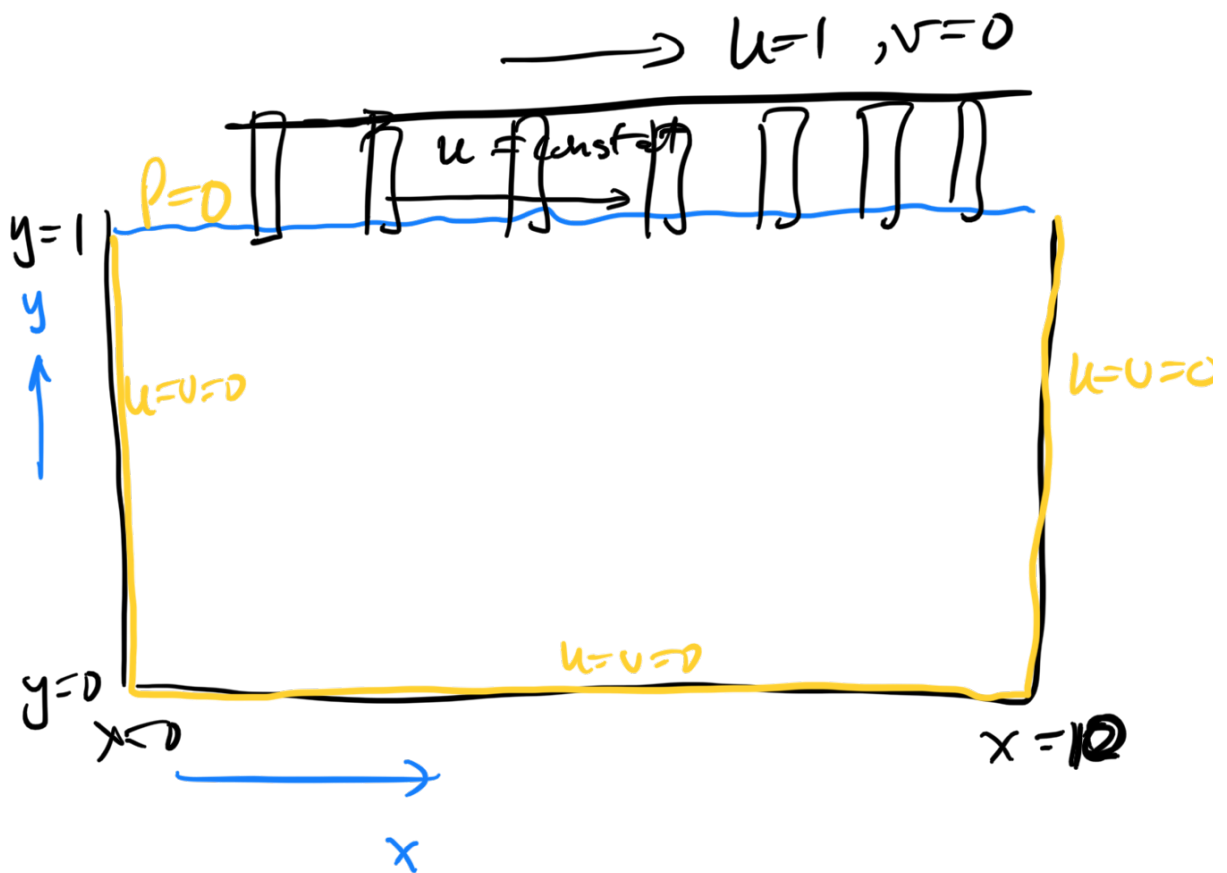
Atomic Clock

$$v = 0.99 c$$

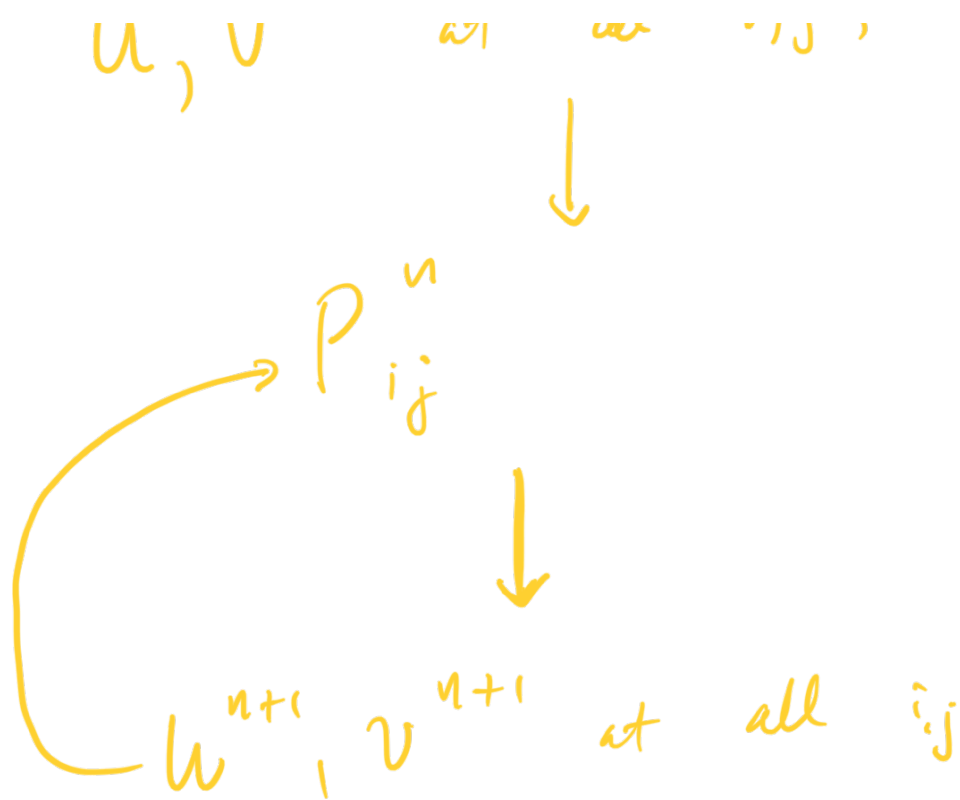
$$\gamma = \frac{1}{\sqrt{1 - 1^2}} = 7.1$$

Movies:

Inter stellar



1. $n=0$ ist die Basis.



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- ENGL 223
 - Foreign Language?
 - MATH MATH 140 or MATH 12
 - AINW + Lab → CHEM 121 + Lab
 - ALGM
-

Kinesiology - Spring of next semester.

Fall 2021

3 FRENCH 102

3 ENGL 223

3 CHEM 121

1 CHEM 121L

3 BIOL 211

1 BIOL 212

[14] MATH

Spring 2022

3 FREN 200

3 CHEM 122

1 CHEM 122L

3 BIOL 212

1 BIOL 212L

[11]

ALGM

MATH 125

pre req for BIO 300

Foreign Language

SPAN 200

ENGL 223

FALL 2021

MATH 247

ENGL 223

CDEW 214

PHYS 303

?

Spring 2022

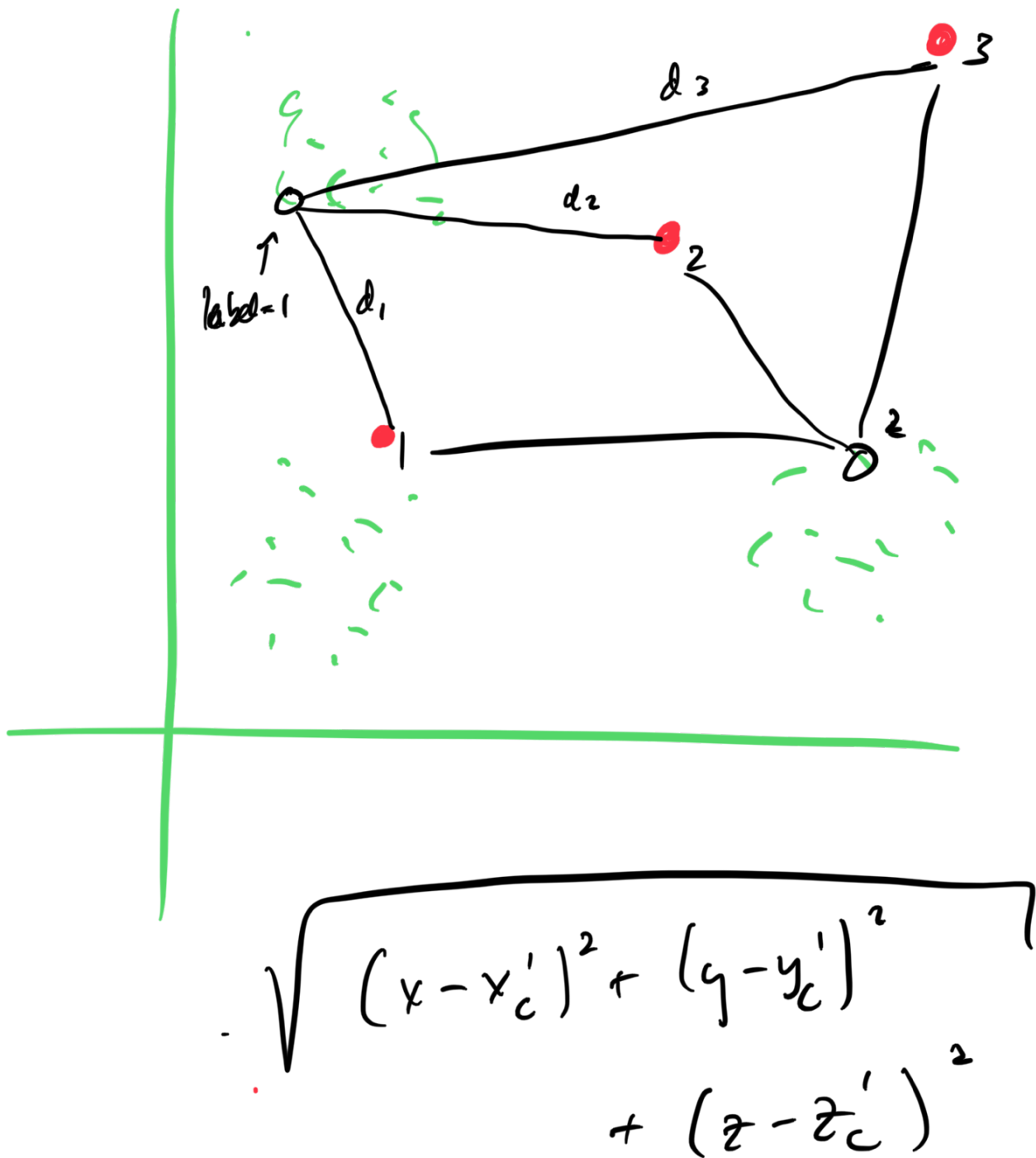
MATH 250

① SPAN 200

② AIDE / AICE / AIGM / AIWT ✓

③

PHYS 341



abc ← numpg array!